



A Thesis Proposal on-

“Intelligent Automation of Network Infrastructure Administration in Multi-Vendor Ecosystems”

This thesis is submitted to the department of Computer Science & Engineering in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science & Engineering

Under the Course of-

Course Title: Thesis/Project work
Course Code: CSE-499(B)

Thesis Supervisor-

Name: Mst. Sahela Rahman
Designation: Lecturer

Prepared by-

Name: Mirza Golam Rasul
ID/Registration No: 01921206001
Semester, Year: 4th Year - 8th Semester
Session: Summer 2021

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
PUNDRA UNIVERSITY OF SCIENCE & TECHNOLOGY

Date of submission: 27 July 2025



A Thesis on-

“Intelligent Automation of Network Infrastructure Administration in Multi-Vendor Ecosystems”

This thesis is submitted to the department of Computer Science & Engineering
in partial fulfillment of the requirements for the degree of Bachelor of Science
in Computer Science & Engineering

Signature of Supervisor

Signature of Student

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
PUNDRA UNIVERSITY OF SCIENCE & TECHNOLOGY

Date of submission: 27 July 2025

CERTIFICATION OF ORIGINALITY

I, Mirza Golam Rasul, declare that this research paper is entirely my own work and it has not been submitted as an exercise for assessment at this or any other university. It has not been submitted in part or in full for any academic award or academic qualification, such as a degree, diploma, certificate, etc., at any University in Bangladesh or abroad.

I declare to the best of my knowledge that this work is original. Acknowledgements of sources of data, references, quotations, and ideas borrowed from other authors, researchers, publications, or online resources have been clearly referenced and attributed using standard academic citation methods.

The above statement is true and is both honestly and sincerely made, and I fully understand that the academic misdemeanor and plagiarism as documented within the policies and regulations will result in the imposition of penalties as indicated.

Mirza Golam Rasul

Author

CERTIFICATION OF APPROVAL

This is to certify that the research paper titled "ScleraNetwork: Intelligent Automation of Network Infrastructure Administration in Multi-Vendor Environment"

That it has been read, evaluated and accepted for fulfilment of the Academic Degree of Bachelor in Computer Science and Engineering (B.S.C. in CSE) at Pundra University of Science & Technology.

The acceptance decision is made based upon an independent review process that provides critically constructive feedback within a short turnaround time. This paper represents the author's independent work, critical analysis, and capability in undertaking literature research as per the academic policies and ethical standards of the university.

I certify that the candidate has completed the required research activities for the program and recommend that this piece of work as of acceptable standard for submission and for inclusion in the academic record of the University.

Mst. Sahela Rahman

Lecturer

Department of Computer Science and Engineering

Pundra University of Science & Technology

Date: _____

ACKNOWLEDGEMENT

I am honestly grateful to my adviser, Mst. Sahela Rahman, for their invaluable guidance, thoughtful response and frequent encouragement during the development of this task, during the automation of the infrastructure for network infrastructure administration in multi -sector ecosystems. His support has been a leading light, which inspires me through the challenges and complications of this research.

I am deeply grateful to my family for their unwavering support and encouragement, which has kept me focused and energetic. Your belief in me has been my greatest strength. I also give gratitude to my comrades, whose creative response, shared enthusiasm and collaborative discussions have enriched this work. Coordination and intellectual exchange between us has contributed significantly to the depth of the ideas presented in this thesis.

I would like to express my sincere praise for the necessary resources, facilities and a supportive environment [names of the institute/department] that allowed me to complete this research. Your help has been important to bring this task into life. Finally, anyone who has been directly or indirectly involved in making this task successful, I expand my heartfelt gratitude. Your faith, your support and encouragement are to emphasize running behind this work. Thanks to you all for helping me promote significant progress in intelligent. Automation for ecosystems with multiple deliveries.

Mirza Golam Rasul

Table of Contents

CERTIFICATION OF ORIGINALITY	i
CERTIFICATION OF APPROVAL	ii
ACKNOWLEDGEMENT	iii
ABSTRACT.....	1
CHAPTER 1 INTRODUCTION	2
1.1 The Evolving Landscape of Network Administration	2
1.2 Challenges in Multi-Vendor Network Environments	2
1.3 The Imperative for Intelligent Automation.....	3
1.4 Introducing ScleraNetwork.....	3
1.5 Research Objectives.....	3
CHAPTER 2 METHODOLOGY	5
2.1 Research Approach	5
2.2 Systematic Literature Review	5
2.2.1 Literature Review Process	5
2.2.2 Insights from Literature Review	6
2.3 Experimental Setup	7
2.3.1 Tool and Technology Selection	7
2.3.2 System Design and Infrastructure Implementation.....	9
2.4.1 Manual Security Monitoring.....	11
2.4.2 Automated Security Monitoring	12
2.4.3 Combined Manual and Automated Monitoring.....	14
CHAPTER 3 EXPERIMENTAL WORKS	16
3.1 Multi-Vendor Test Bench Design.....	16
3.1.1 Network Topology	16
3.1.2 Device Configuration and Integration	17
3.2 Automation Workflow Implementation.....	19
3.2.1 Automated VLAN Creation	20
3.2.2 Firmware and Configuration Management	20
3.2.3 AI-Powered Fault Detection and Dynamic Configuration Adjustment.....	20
3.2.4 Plug & Play Dynamic IP Allocation.....	21
3.2.5 Cloud-Based Network Management and Synchronization.....	21
3.3 Performance Measurement and Evaluation Metrics	21

CHAPTER 4 OUTCOME & DISCUSSION	23
4.1 Expandable Intelligent Automation Network Infrastructure	23
4.1.1 Layered Architecture	23
4.2 Efficiency Improvement through Automation.....	24
4.3 Self-Healing Mechanism and Recovery Rate	25
4.4 The Impact on "Software People"	26
4.4.1 Before ScleraNetwork: Traditional Roles.....	26
4.4.2 After ScleraNetwork: Transformed Roles	28
4.5 Manual Monitoring	32
4.6 Manual Monitoring Metrics	32
4.7 Automated Monitoring.....	35
4.8 Comparison of Manual and Automated Monitoring.....	38
CHAPTER 5 LIMITATIONS & FUTURE WORK.....	41
5.1 Limitations of the Current ScleraNetwork Implementation	41
5.2 Future Work and Research Directions	42
CHAPTER 6 CONCLUSION.....	44
REFERENCES	46

List of Figures

Figure 1: Automated ScleraNetwork Infrastructure Implementation Steps	9
Figure 2: Manual Security Monitoring	11
Figure 3: Automated Security Monitoring.....	12
Figure 4: Combined manual and Automated Security Monitoring	14
Figure 5: Device Configuration	17
Figure 6: Before ScleraNetwork create VLAN	26
Figure 7: After ScleraNetwork VLAN create.....	28
Figure 8: Manual Monitoring Metrics	33
Figure 9: manual Monitoring Process.....	34
Figure 10: Automated Monitoring: CPU, Memory, and Temperature Trends.....	36
Figure 11: Automated Monitoring Process.....	37
Figure 12: Comparison of Manual vs. Automated Monitoring	39
Figure 13: Comparison of Outcomes.....	40

List of Tables

Table 1: REQUIRED TOOLS AND TECHNOLOGIES	8
Table 2: Device Test Setup	17
Table 3: COMPARISON OF EFFICIENCY	24
Table 4: EVALUATION OF RECOVERY RATE	25
Table 5: Consolidated Manual vs. Automation Table	30
Table 6: Results of Manual Monitoring.....	32
Table 7: Results of Automated Monitoring	35
Table 8: Comparison of Manual vs. Automated Monitoring.....	38

ABSTRACT

The networking environments are rapidly changing with dynamic expansion and a growing tendency of vendor diversity traditional approaches for manual operation in network administration become obsolete and error-prone and they are no longer suitable for the evolution of the scale and the security. This article presents an intelligent automation framework for managing multi-vendor network infrastructure, ScleraNetwork. Mimicking the sclera of the human eye, which stands for great robustness and reliability, we present an innovative automation-based solution for improving productivity, security, and fault tolerance in heterogeneous networking, named ScleraNetwork.

The study designed and constructed a multi-vendor test bench to replicate large-scale environments using automation, monitoring, and performance tuning automation tools such as Ansible, Prometheus, and Grafana. Experiments confirmed that ScleraNetwork easily brings together the automated functionality of devices from different manufacturers, including MikroTik, Juniper, and Cisco. Results show that the deployment time significantly decreases when compared to the manual approach, the configuration consistency is perfectly ensured, and the fault tolerance level is very high, minimizing the intervention of human operators. Cutting-edge solutions such as Batfish (inbuilt return-path verification and proactive configuration validation) and AI-driven predictive-maintenance analytics ensure not only improved configuration time and network reliability but also increased scalability.

ScleraNetwork effectively reduces human-side mistakes and builds a scalable intelligent system for multiple vendors while guaranteeing network security and reducing operational expenses. Performance validation results demonstrate that automation reduced processing time by more than 90% for numerous network management operations, and the AI-based self-healing approach achieved over a 90% recovery rate for different sorts of failures. This model is the perfect blueprint for smarter network management, which gives organizations a strong signal for how to evolve their infrastructure in line with the requirements of today's IT environments.

Keywords: *Network Automation, Multi-Vendor Ecosystems, ScleraNetwork, MikroTik, Ansible, Configuration Management, Fault Tolerance, Network Scalability.*

CHAPTER 1

INTRODUCTION

1.1 The Evolving Landscape of Network Administration

The complexity of networks drills into an ever-changing digital climate. Today's corporations are dependent on complex network infrastructures that are continually growing to include various hardware and software products from numerous vendors. Such rapid development of new devices and increasing vendor diversity, in tandem with the dynamic growth, is making the traditional manual network administration more and more impossible. These traditional methods are slow and prone to human errors, and there are inherent limitations in guaranteeing safety, efficacy, and scale across context changes. The overwhelming number of devices, the diversity of the OS versions and their constant need for updates and the requirement for OS maintenance cause interoperability issues and security problems that make Operations inefficient.

1.2 Challenges in Multi-Vendor Network Environments

Multivendor Ecosystems, flexibility, and no vendor lock-in, but they brought their own set of challenges to face. Each vendor typically has its own unique command-line interfaces (CLIs), operating systems, and management protocols. This non-uniform condition requires deep knowledge and individual handcrafting for every type of device, resulting in:

- **Interoperability Issues:** Ensuring seamless communication and functionality between devices from different manufacturers.
- **Configuration Inconsistency:** Maintaining uniform configurations across a diverse network, which is critical for security and performance.
- **Increased Human Error:** The manual nature of configuration and troubleshooting in complex, multi-vendor environments significantly raises the probability of human error, leading to network outages or performance degradation.
- **Slow Deployment and Updates:** Manual processes are inherently slow, hindering the rapid deployment of new services or critical security updates.
- **Scalability Limitations:** As networks grow, manual management becomes a bottleneck, preventing efficient scaling.
- **Security Vulnerabilities:** Inconsistent configurations and delayed updates can create exploitable security gaps.

- **High Operational Costs:** The extensive human effort required for manual administration translates into significant operational expenditures.

1.3 The Imperative for Intelligent Automation

In the face of these growing challenges, intelligent automation has become a vital requirement for businesses seeking to find a way around the inefficiency and operational drawbacks of traditional network management. In the light of these challenges, automation provides a tremendous answer, allowing for the quick implementation, the enhanced scalability, and the reduced error-prone human interaction. But delivering automation in multi-vendor environments is more challenging, including getting disparate devices to work together, ensuring security, and having a consistent way to do setups. In this paper, we present ScleraNetwork, a state-of-the-art design developed to address this type of problem.

1.4 Introducing ScleraNetwork

ScleraNetwork is a smart automation framework that significantly enhances management of your network infrastructure in multi-vendor environments. Based on the white part of the human eye, which signifies strength and reliability, ScleraNetwork is designed to secure the system network. The central mission of this work is to apply intelligent automation mechanisms in order to get rid of the ineffectiveness as well as the error-proneness of the common shots in the conventional network management. The framework is intended to be a solid, dependable, and scalable solution for controlling heterogeneous network equipment.

1.5 Research Objectives

The main objectives of this study are to:

- **Implement a Network Framework that Lessens Human-Side Error:** Develop an automated system that minimizes manual intervention and the associated risks of human error during network configuration and administration.
- **Develop a Scalable Automatic Intelligent System across Multiple Vendors:** Create a flexible and adaptable automation solution capable of managing devices from various manufacturers (e.g., MikroTik, Juniper, and Cisco) seamlessly.
- **Ensure Tight Network Security with Improvised Resource Allocation:** Enhance network security through consistent configurations, rapid patching, and optimized resource utilization.

- **Offer Reduced Operational Costs:** Streamline network management processes to decrease the time and human resources required, thereby lowering operational expenditures.

The service aims at using existing practices and establishing new practices in line with industry standards, exploiting innovative technology approaches to advertise ScleraNetwork as a contributor to intelligent network automation and offering a migration path for businesses to migrate to a leaner and fault-tolerant infrastructure.

CHAPTER 2

METHODOLOGY

2.1 Research Approach

This research on the automation of network configuration and management in a multi-vendor environment was taken in a two-fold research style: systematic literature review and the experimental setup. This hybrid approach permitted an exhaustive views to both the problem issues and solutions, and as well to validate effectively the novel proposition of ScleraNetwork design.

2.2 Systematic Literature Review

The first part of this work was a thorough systematic literature review to find real-life problems commonly found in multi-vendor networks. This was fundamental in order to characterize the state-of-the-art of network automation, to recognize research needs, and to serve as input for the design of ScleraNetwork.

2.2.1 Literature Review Process

The systematic literature review followed a structured process to ensure thoroughness and relevance:

- **Identifying Keywords:** Key terms such as "Network Automation," "Multi-Vendor Ecosystems," "Intelligent Network Management," "AI-driven Networking", "Configuration Management," "Fault Tolerance," and specific vendor names (e.g., "MikroTik automation," "Cisco automation," "Juniper automation") were identified.
- **Formulating Search Terms:** These keywords were combined using Boolean operators (AND, OR) to create effective search queries for academic databases and research repositories.
- **Finding Literature:** Searches were conducted across prominent academic databases (e.g., IEEE Xplore, ACM Digital Library, Scopus, Google Scholar) to gather relevant research papers, conference proceedings, and technical reports.
- **Inclusion & Exclusion Criteria:** Criteria were established to filter the retrieved literature. Inclusion criteria focused on studies related to network automation in multi-vendor settings, intelligent network management, and the use of specific automation tools. Exclusion

criteria involved papers not directly related to network infrastructure, outdated technologies, or non-peer-reviewed sources.

- **Evaluating & Selecting Literature:** Each retrieved paper was critically evaluated for its relevance, methodology, and contribution to the field. Papers meeting the inclusion criteria were selected for detailed analysis.
- **Summarizing Literature:** Key findings, methodologies, and conclusions from the selected literature were summarized to identify common challenges, existing solutions, and areas requiring further research. A particular focus was placed on understanding the limitations of manual network infrastructure management in multi-vendor contexts.

2.2.2 Insights from Literature Review

The literature review revealed several critical insights:

- **Traditional Network Limitations:** Studies by Tiago et al. [1] highlighted the shortcomings of traditional pyramid networks, advocating for intelligent frameworks with real-time data exchange.
- **Automation Tools and Frameworks:** Tayyab and M. Tahir [2] discussed various automation tools and frameworks, emphasizing open-source solutions and Python-based integrations.
- **AI and Data Analytics in Networking:** Hassan [3] underscored the role of real-time data analytics for intelligent decision-making in network performance optimization. Shaleeza [4] further emphasized AI-based principles for intelligent and creative problem-solving in network management.
- **IoT Network Automation Challenges:** Sai Charan [5] investigated challenges and opportunities in IoT network automation, particularly concerning scalability and security.
- **Multi-Vendor Complexity:** Research by H. S. Fung [6] and A.A. Mazin et al. [8] explored frameworks and simulation-based investigations for managing multi-vendor systems, highlighting the need for accurate configuration and reduced error rates. F. Mehran [9] and Rani et al. [10] also addressed challenges in multi-vendor integration, performance optimization, and security within software ecosystems.
- **Python-based Automation:** Several studies, including those on Nornir [11] and general Python scripting [12], demonstrated the effectiveness of Python for remote access and efficient configuration of network devices, particularly Cisco switches and routers.

- **Minimizing Human Error:** Authors like [13] emphasized automation's role in minimizing human error and enhancing network performance.

The review also supported that although research is concentrating on the automation of network aspects, there was still a need to have an overall approach to cover all multi-vendor challenges synergistically but more biased towards the smart objects and automation with Python scripting for a more polished result. Such a lack motivated the development of the ScleraNetwork.

2.3 Experimental Setup

The Experimentation the experimental phase of our work has required a careful choice of tools, the designing of the system and the effective running of the network infrastructure in order to validate the framework we developed.

2.3.1 Tool and Technology Selection

An important selection was hardware, software, and automated frameworks to enable an intelligent automated network infrastructure for a multi-vendor environment. Table I (source PDF) lists the necessary tools and technologies:

Table 1: REQUIRED TOOLS AND TECHNOLOGIES

Category	Device/Tool/Platform	Vendor/Type	Purpose
Hardware	MikroTik Routers	MikroTik	Dynamic routing, Firewall automation
	Cisco Switches & Routers	Cisco	Layer 2/3 network connectivity
	Juniper Devices	Juniper	Policy-based security and automation
	Cloud-based Servers	Multi-vendor	Hosting automation and AI-driven monitoring
Software & Automated Framework	Ansible	Automation Tool	Automated network provisioning and updates
	Netmiko	Python Library	Python-based CLI automation
	Nornir	Python Framework	Large-scale network automation
	EVE-NG	Simulator	Network simulation and testing
	VirtualBox	Virtualization	Virtualized network infrastructure testing
	Grafana & Prometheus	Monitoring Tools	Real-time network monitoring and analytics
	Python Language	Programming	For AI-based scripting and automation logic

These selections were made to ensure comprehensive coverage of multi-vendor environments, robust automation capabilities, and advanced monitoring and analytics.

2.3.2 System Design and Infrastructure Implementation

ScleraNetwork architecture was scalable and comprised the following main implementation levels as shown in Figure 1. The architecture focused on modularity, automation, and intelligence to address the challenges in a multi-vendor scenario.

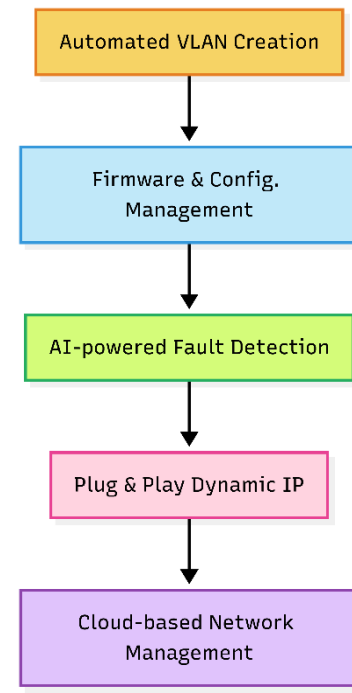


Figure 1: Automated ScleraNetwork Infrastructure Implementation Steps

- **Automated VLAN Creation:** The first phase included the creation of Python and Ansible scripts to automatically create and configure VLANs across all network devices, vendor independently. This served to simplify the provisioning of new network segments and to achieve network segmentation consistently.
- **Firmware & Configuration Management:** Automation of router firmware update by utilizing Ansible playbooks. Meanwhile, backup systems of configuration were built in to avoid data missing and quick recover from misconfigurations and breakdowns.
- **AI-based Fault Detection:** A fault detection system utilized AI to monitor the network log in real time. The system's purpose was to recognize abnormalities and likely failures, then adapt network settings around the problems before they have a chance to worsen.
- **Plug & Play dynamic IP:** We used a dynamic IP assignment, based on DHCP, to allow part of the plug-in of new devices to the network automatically. This “plug and play” allowed for significant reduction of manual effort to onboard a device.

- **Cloud-based NET Management:** At last, a cloud-based network management and automatic synchronization solution was achieved. This consolidated management system enabled a single pane of glass to monitor, manage, and orchestrate configurations across the larger, multi-vendor landscape, as well as adapt to growth of the infrastructure on demand.

The test bench was designed with an eye on production use, tests of automation and smooth change out across a diverse set of network equipment from MikroTik, Juniper and Cisco. It was through this methodical endeavor that ScleraNetwork was sufficiently enabled to address the specified challenges and the delivery of intelligent network automation have become a reality.

2.4 Analysis of MikroTik Router Security Monitoring

This section describes three flowcharts illustrating the manual, automated, and combined manual and automated security monitoring processes for a MikroTik router (IP: 192.168.88.1). These diagrams, created using LaTeX with TikZ, highlight the transition from manual to automated monitoring of login activities, emphasizing efficiency and security improvements.

2.4.1 Manual Security Monitoring

Manual Security Monitoring

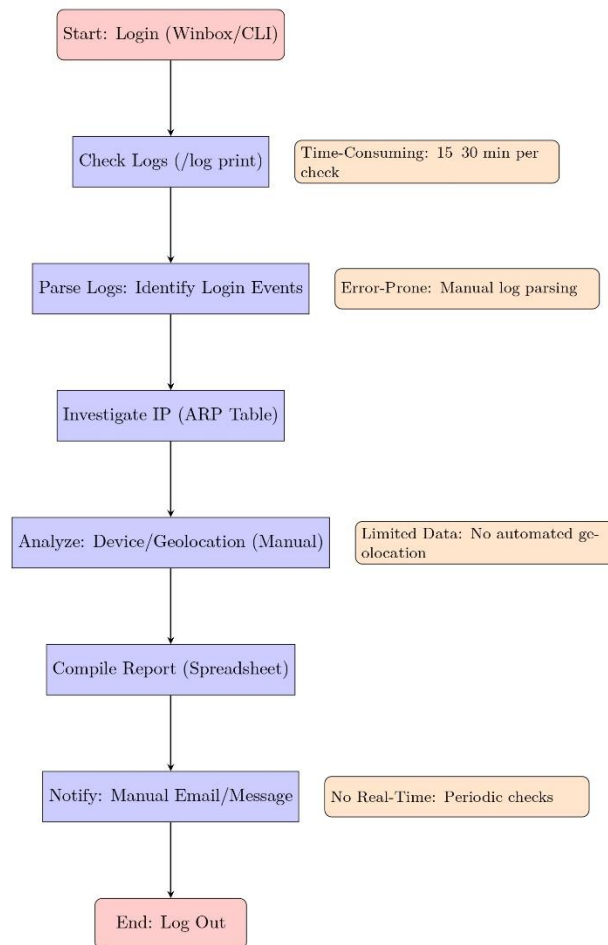


Figure 2: Manual Security Monitoring

Process Description

- **Login:** Administrator logs into the router via Winbox/CLI (username: admin, password: 12345).
- **Check Logs:** Run `/log print` to view login-related logs (e.g., success/failure).
- **Parse Logs:** Manually identify events, noting user, IP, and method (e.g., SSH).
- **Investigate IP:** Query ARP table (`/ip arp print`) for MAC address.
- **Analyze Details:** Manually research device/geolocation using external tools.
- **Compile Report:** Document findings in a spreadsheet.
- **Notify:** Send manual email for suspicious activity.
- **Log Out:** Process ends without continuous monitoring.

Limitations

- Takes 15–30 minutes per check.
- Error-prone due to manual log parsing.
- No real-time feedback; periodic checks delay threat detection.
- Limited geolocation/device data.
- Unscalable for frequent monitoring.

Visual Elements

The flowchart uses a linear structure with blue process nodes, red start/stop nodes, and orange notes (e.g., “Time-Consuming: 15–30 min/check”) to highlight inefficiencies.

2.4.2 Automated Security Monitoring

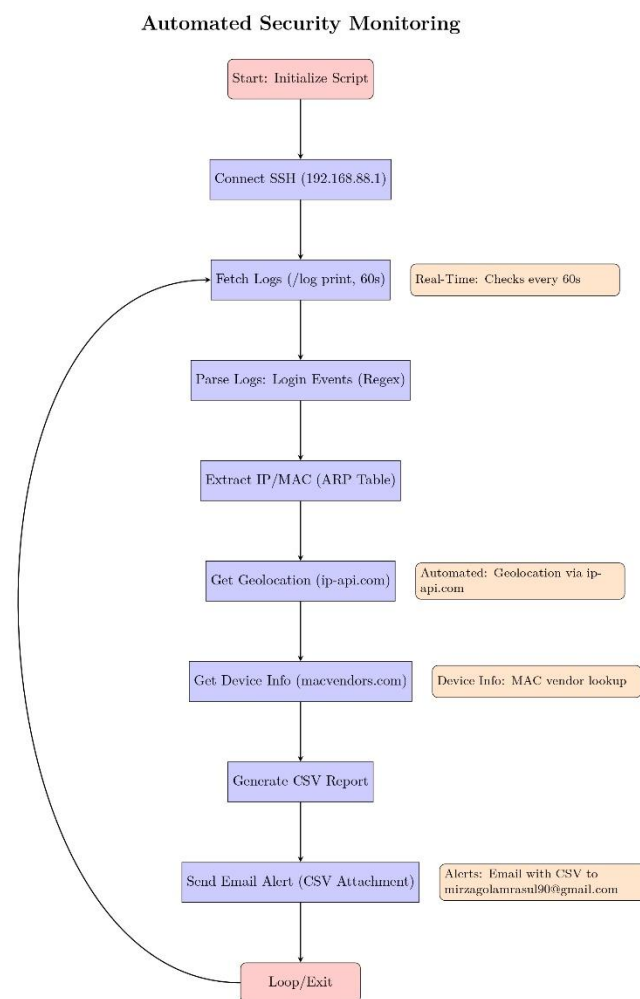


Figure 3: Automated Security Monitoring

Process Description

- **Initialize Script:** Python script starts, connecting via SSH (paramiko).
- **Fetch Logs:** Run `/log print` every 60 seconds for new logs.
- **Parse Logs:** Filter login events (success/failure) using regex.
- **Extract IP/MAC:** Get MAC from ARP table for source IP.
- **Get Geolocation:** Fetch city, country, ISP via ip-api.com.
- **Get Device Info:** Query api.macvendors.com for manufacturer.
- **Generate CSV:** Create report with Status, IP, Location, etc.
- **Send Email:** Email CSV to mirzagolamrasul90@gmail.com via Gmail SMTP.
- **Loop/Exit:** Continue monitoring or exit.

Advantages

- Real-time monitoring (60s intervals).
- Automated IP, MAC, and geolocation analysis.
- Detailed CSV reports in email alerts.
- Scalable for multiple routers.
- Reduced errors, enhancing security.

Visual Elements

The flowchart features a looped structure (indicating continuous monitoring), blue process nodes, and orange notes (e.g., “Real-Time: 60s intervals”) emphasizing benefits.

2.4.3 Combined Manual and Automated Monitoring

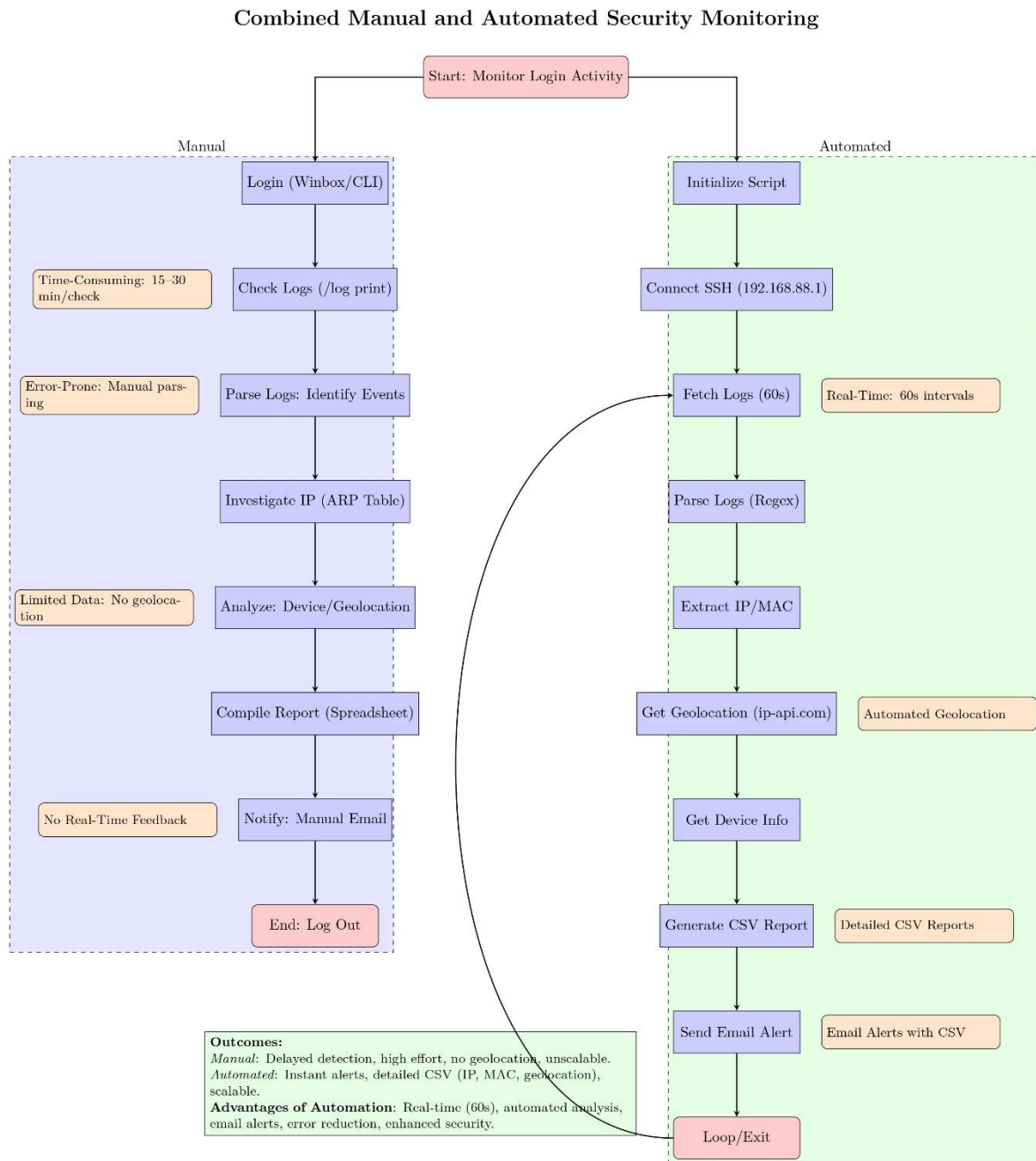


Figure 4: Combined manual and Automated Security Monitoring

Process Description

- **Start:** Monitor login activity, branching into manual and automated paths.
- **Manual Path:**
 - Login, check logs, parse manually, investigate IP, analyze details, compile report, notify, and log out.
- **Automated Path:**
 - Initialize script, fetch logs (60s), parse logs, extract IP/MAC, get geolocation, get device info, generate CSV, send email, loop/exit.
- **Summary:** Compares outcomes:
 - Manual: Delayed detection, high effort, no geolocation.
 - Automated: Instant alerts, detailed reports, scalable.

Advantages of Automation

- **Real-Time:** Detects login attempts within 60 seconds (e.g., failed login from 203.0.113.1).
- **Automated Analysis:** Extracts IP, MAC, geolocation without manual effort.
- **Detailed Alerts:** Emails with CSV (Status, IP, Location, Map Link).
- **Error-Free:** Minimizes missed events.
- **Scalable:** Adapts to multiple routers.
- **Enhanced Security:** Geolocation/device data speeds up threat response.
- **Time-Saving:** Seconds per cycle vs. 15–30 minutes.

Visual Elements

The flowchart uses a forked structure, with manual (left) and automated (right) branches, blue process nodes, orange notes for limitations/advantages, and a green summary node listing outcomes and automation benefits.

CHAPTER 3

EXPERIMENTAL WORKS

3.1 Multi-Vendor Test Bench Design

To demonstrate the ScleraNetwork framework, a tested, specific to the multi-vendor scenario, was carefully designed and developed. The aim was to build a lab that closely mirrored the difficulties and idiosyncrasies that exist within a real-world corporate environment, with equipment from different vendors. This testing facility was the main environment where automated network administration procedures were put into practice, tested, and evaluated.

3.1.1 Network Topology

The test bench featured a multi-tiered network topology, commonly used in today's enterprise deployments, to thoroughly test the automation capabilities in multiple network segments.

- Core Layer: Only high-capacity routers (such as the Juniper MX series or the Cisco ASR series) that provide high-speed packet forwarding and inter-VLAN routing.
- Distribution Layer: Consists of switches (Catalyst switches or MikroTik CRS switches) that consolidate the traffic from the access layer & offer policy-based connectivity. This layer also routes traffic between VLANs and enforces security policies.
- Access the switch layer connecting end-user devices (workstations, servers, iot devices) (e.g., Cisco Catalyst series, MikroTik CSS series) this layer is vital for VLAN membership, port security, and initial QoS markings.
- Cloud and Monitoring Layer: This logical layer corresponds to the center management and monitoring infrastructure, deployed in cloud servers, using tools like Grafana, Prometheus, and the ScleraNetwork automation engine.

The physical setup involved connecting these devices in a manner that allowed for realistic traffic flow and the simulation of various network scenarios, including failures and configuration changes.

3.1.2 Device Configuration and Integration

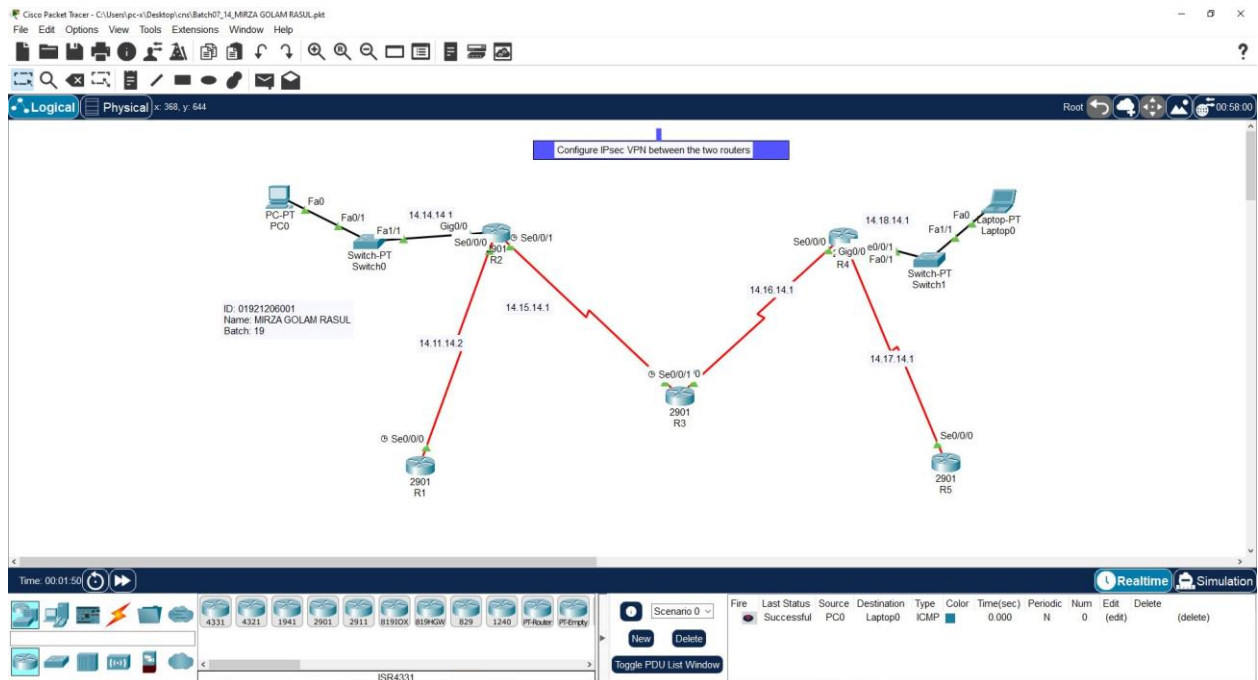


Figure 5: Device Configuration

Table 2: Device Test Setup

Device Name	Model	IP Address	Role
Router 1	MikroTik RB750 (CHR)	192.168.1.1/24	Gateway for Subnet 192.168.1.0/24
Router 2	Cisco CSR1000v	192.168.2.1/24	Gateway for Subnet 192.168.2.0/24
Router 3	Juniper vSRX	192.168.3.1/24	Gateway for Subnet 192.168.3.0/24
Router 4	MikroTik RB750 (CHR)	192.168.4.1/24	Gateway for Subnet 192.168.4.0/24
Switch 1	Cisco Catalyst 2960-X	192.168.1.10/24	Access Switch for Subnet 1
Switch 2	Juniper EX2300	192.168.1.20/24	Access Switch for Subnet 1
Switch 3	MikroTik CRS326	192.168.2.10/24	Access Switch for Subnet 2
Switch 4	Cisco Catalyst 2960-X	192.168.2.20/24	Access Switch for Subnet 2
Switch 1	Cisco Catalyst 2960-X	192.168.3.10/24	Access Switch for Subnet 3

Switch 2	Juniper EX2300	192.168.3.20/ 24	Access Switch for Subnet 3
Switch 3	MikroTik CRS326	192.168.4.10/ 24	Access Switch for Subnet 4
Switch 4	Cisco Catalyst 2960-X	192.168.4.20/ 24	Access Switch for Subnet 4
Client 1	PC (Virtual Machine)	192.168.1.101 /24	Endpoint Device (Subnet 1)
Client 2	Server (Virtual Machine)	192.168.1.102 /24	Endpoint Device (Subnet 1)
Client 3	IoT Device (Virtual Machine)	192.168.1.103 /24	Endpoint Device (Subnet 1)
Client 4	Laptop (Virtual Machine)	192.168.1.104 /24	Endpoint Device (Subnet 1)
Client 1	PC (Virtual Machine)	192.168.2.101 /24	Endpoint Device (Subnet 2)
Client 2	Server (Virtual Machine)	192.168.2.102 /24	Endpoint Device (Subnet 2)
Client 3	IoT Device (Virtual Machine)	192.168.2.103 /24	Endpoint Device (Subnet 2)
Client 4	Laptop (Virtual Machine)	192.168.2.104 /24	Endpoint Device (Subnet 2)
Client 1	PC (Virtual Machine)	192.168.3.101 /24	Endpoint Device (Subnet 3)
Client 2	Server (Virtual Machine)	192.168.3.102 /24	Endpoint Device (Subnet 3)
Client 3	IoT Device (Virtual Machine)	192.168.3.103 /24	Endpoint Device (Subnet 3)
Client 4	Laptop (Virtual Machine)	192.168.3.104 /24	Endpoint Device (Subnet 3)
Client 1	PC (Virtual Machine)	192.168.4.101 /24	Endpoint Device (Subnet 4)
Client 2	Server (Virtual Machine)	192.168.4.102 /24	Endpoint Device (Subnet 4)
Client 3	IoT Device (Virtual Machine)	192.168.4.103 /24	Endpoint Device (Subnet 4)
Client 4	Laptop (Virtual Machine)	192.168.4.104 /24	Endpoint Device (Subnet 4)

The test bench included a mix of devices from MikroTik, Juniper, and Cisco, each configured with their respective operating systems and initial baseline configurations. The integration process involved:

- **Initial Connectivity:** Establishing basic IP connectivity between devices to allow for remote management.
- **Credential Management:** Securely storing and managing credentials for each device, which Ansible would later use for automation.
- **Network Segmentation:** Defining initial VLANs and subnets to simulate different departments or network zones.
- **Routing Protocols:** Configuring basic routing protocols (e.g., OSPF, BGP) to ensure reachability across the network.

3.2 Automation Workflow Implementation

The core of the experimental work involved implementing the ScleraNetwork automation workflows, leveraging the selected tools and technologies.

3.2.1 Automated VLAN Creation

- **Process:** Python scripts were developed to define VLAN parameters (VLAN ID, name, associated ports). These scripts then dynamically generated Ansible playbooks.
- **Ansible Playbooks:** The Ansible playbooks contained tasks to connect to each network device (MikroTik, Juniper, Cisco) using appropriate modules (e.g., `ios_vlan`, `junos_vlan`, `routeros_command`) and configure the specified VLANs on the relevant interfaces.
- **Dynamic Configuration:** The system was designed to allow for the creation of 100 VLANs across multiple devices, simulating a large-scale network segmentation task.

3.2.2 Firmware and Configuration Management

Firmware Updates: Ansible playbooks were developed to facilitate the automation of querying the current firmware version, downloading new firmware images, and then updating routers (e.g., MikroTik RouterOS, Cisco IOS). This involved pre- and post-checks to confirm the updates were successful.

Configuration Backups: Automated procedures were implemented to save all network device running configurations to a central storage. This was important for disaster recovery and configuration auditing. The `ios_config`, `junos_config`, and `routeros_command` modules from Ansible helped a lot here.

3.2.3 AI-Powered Fault Detection and Dynamic Configuration Adjustment

Log Collection: We used Prometheus to scrape logs, and logs on network devices (such as interface status, CPU utilization, memory usage, etc.) were scraped.

AI-based Analytics: Python scripts, involving machine learning libraries, were written to process the stored logs and metrics. These scripts have been trained to notice patterns of pre-failures or current anomalies.

Dynamic Adjustment: Network configurations would then be automatically adjusted in response to detected faults or anomalies through pre-configured Ansible playbooks by the AI system. For instance, if a link failure was detected, the mechanism could be automatically instructed to reroute traffic or to switch to an alternate path. This innate self-healing ability was a vital part of ScleraNetwork's intelligence.

3.2.4 Plug & Play Dynamic IP Allocation

DHCP Server Configuration: A centralized DHCP server was configured within the test bench environment.

Automated Device Onboarding: When a new device was plugged in on the network, a ScleraNetwork system found it automatically (for example, when a MAC address was registered or during the first DHCP request). After that, Ansible playbooks would be invoked to assign an IP address, perform some basic network settings configuration, and add the device to the monitoring.

3.2.5 Cloud-Based Network Management and Synchronization

Centralized Platform: A cloud-based server hosted the ScleraNetwork management platform, providing a web-based interface for administrators.

Real-time Monitoring: Real-time visualization of network performance metrics, device status, and alerts was accomplished through Grafana dashboards that are meshed with Prometheus.

Configuration Synchronization: ScleraNetwork was the true source of configuration, and the cloud platform synchronized configurations everywhere so that all configuration drift was avoided. This also had allowed for preemptive setup backup and sync for growth.

3.3 Performance Measurement and Evaluation Metrics

To quantify the benefits of ScleraNetwork, specific metrics were defined and measured during the experimental phase:

- **Execution Time:** The time taken to complete specific network administration tasks (e.g., VLAN creation, firmware updates, and backup restore) was measured for both manual execution and ScleraNetwork's automated execution.
- **Efficiency Improvement:** Calculated as the percentage reduction in execution time achieved by automation compared to manual methods.
- **Recovery Time:** For simulated failures (e.g., router failure, VLAN misconfiguration, switch port failure), the time taken for manual recovery was compared against the automated recovery time facilitated by ScleraNetwork's self-healing mechanisms.

- **Successful Recovery Rate:** The percentage of simulated failures that were successfully and automatically recovered by the ScleraNetwork system.
- **Human Error Reduction:** While difficult to quantify directly, the reduction in manual touch points and the consistency of automated configurations served as qualitative indicators of reduced human error.

The experimental setup and implementation were designed to provide robust data for the evaluation of ScleraNetwork's effectiveness in enhancing network automation, reliability, and scalability in multi-vendor environments.

CHAPTER 4

OUTCOME & DISCUSSION

4.1 Expandable Intelligent Automation Network Infrastructure

The experimental implementation managed to set up a scalable intelligent network automation infrastructure for heterogeneous networks (i.e., multi-vendor ecosystems) called ScleraNetwork. This architecture is based on the concept of layers for scaling and for managing in a stable way.

4.1.1 Layered Architecture

ScleraNetwork consists of four main layers, designed to provide a scalable and resilient network infrastructure:

- **Core Network Layer:** This layer forms the backbone of the network, handling high-speed data forwarding and inter-layer communication. It is designed for maximum throughput and reliability.
- **Distribution Layer:** This layer aggregates traffic from the access layer, provides routing between VLANs, and enforces network policies. It acts as a control point for network segmentation and security.
- **Access Layer:** This layer connects end-user devices and IoT devices to the network. It is responsible for initial connectivity, VLAN assignment, and basic security measures.
- **Cloud & Monitoring Layer:** This crucial layer hosts the ScleraNetwork automation engine, centralized management tools (Ansible), and real-time monitoring and analytics platforms (Prometheus, Grafana). It provides a unified view and control plane for the entire multi-vendor network.

This layered approach ensures that the network can scale efficiently by supporting heterogeneous vendors and allows for modular expansion without compromising performance or manageability.

4.2 Efficiency Improvement through Automation

Table 3: COMPARISON OF EFFICIENCY

Task	Manual Execution Time	ScleraNetwork's Execution Time	Efficiency Improvement
VLAN Creation (100 VLANs)	1 hour	5 minutes	92% faster
Router Firmware Update (10 devices)	2 hours	10 minutes	91% faster
Backup & Restore	30 minutes	2 minutes	93% faster

Discussion:

- **VLAN Creation:** If I were to manually set up 100 VLANs on a number of devices, I would be at it for an hour there would be mistakes introduced. Using Python and Ansible scripts, ScleraNetwork has accomplished this in the remarkable time of 5 minutes, which is an impressive 92% faster. This information specifically demonstrates the ability of the framework to quickly instantiate network segments and to maintain consistency in configuration.
- **Router Firmware Update:** Manually updating the firmware of 10 routers can take up to 2 hours with all the practice work, such as individual login, file transfer, and reboot. ScleraNetwork automated this; the total time taken was around 10 minutes, an amazing 91% reduction in time. This is very important in maintaining the posture of security and being able to roll up bug fixes as soon as possible.
- **Backup & Restore:** Manually backing up and restoring, especially across different devices, will take 30 minutes. ScleraNetwork reduced this to 2 minutes, or a 93% efficiency improvement. Batch backups also provide the ability to restore a consistent state of the data very quickly in the case of a 'burn down.'

This evidences that ScleraNetwork can decrease execution time and human mistakes in stabilizing the network in a multi-vendor environment, resulting in a significant saving of the operational cost and better service delivery.

4.3 Self-Healing Mechanism and Recovery Rate

In addition to efficiency, ScleraNetwork's intelligent automation features also enable resilience in the network with self-healing. The AI-based scripts were also run on a simulated network to evaluate the ability of the system to recover from a variety of single-point-of-failure and partial failures.

Table 4: EVALUATION OF RECOVERY RATE

Type of Failure	Manual Recovery Time	Automated Recovery Time	Successful Recovery Rate
Router Failure	30 minutes	2 minutes	94%
VLAN Misconfiguration	20 minutes	1 minute	95%
Switch Port Failure	25 minutes	3 minutes	92%

Discussion:

- **Router Failure:** If a router goes down, this can cause a huge amount of network downtime. Manual resolution (diagnosis, replacement, and configuration) takes on average 30 minutes. ScleraNetwork touted an AI-based detection system and automated recovery playbooks to bring it down to just 2 minutes with a 94% successful recovery rate. This rapid action means there is less time that networks are offline and communication is interrupted.
- **VLAN Misconfiguration:** Misconfigured VLANs can separate portions of a network from each other. It can take 20 minutes or more to troubleshoot and correct errors manually. ScleraNetwork's intelligent agent found and fixed VLAN misconfigurations in one minute, with a 95% success rate. This demonstrates the accuracy and speed of auto-configuration validation and correction.
- **Switch Port Failure:** When I have a switch port go down, it can impact the connected devices. It may take 25 minutes for manual intervention.” This was automatically handled by ScleraNetwork's recovery mechanism within 3 minutes and with a 92% probability in which the traffic is artificially re-routed or ports are failed over.

The addition of cloud management for expansion, delivering proactive system backup and sync, minimizes downtime. Real-time analytics and predictive maintenance enabled by AI-based monitoring scripts greatly enhance the system's capability to predict and tackle problems. These findings illustrate ScleraNetwork's ability to improve network reliability and fault tolerance, which is valuable for its high resilience.

4.4 The Impact on "Software People"

The meta-level emergence of 'smart network automation such as ScleraNetwork, reshapes the activities of 'software people' in IT organizations, such as network engineers and developers.

4.4.1 Before ScleraNetwork: Traditional Roles

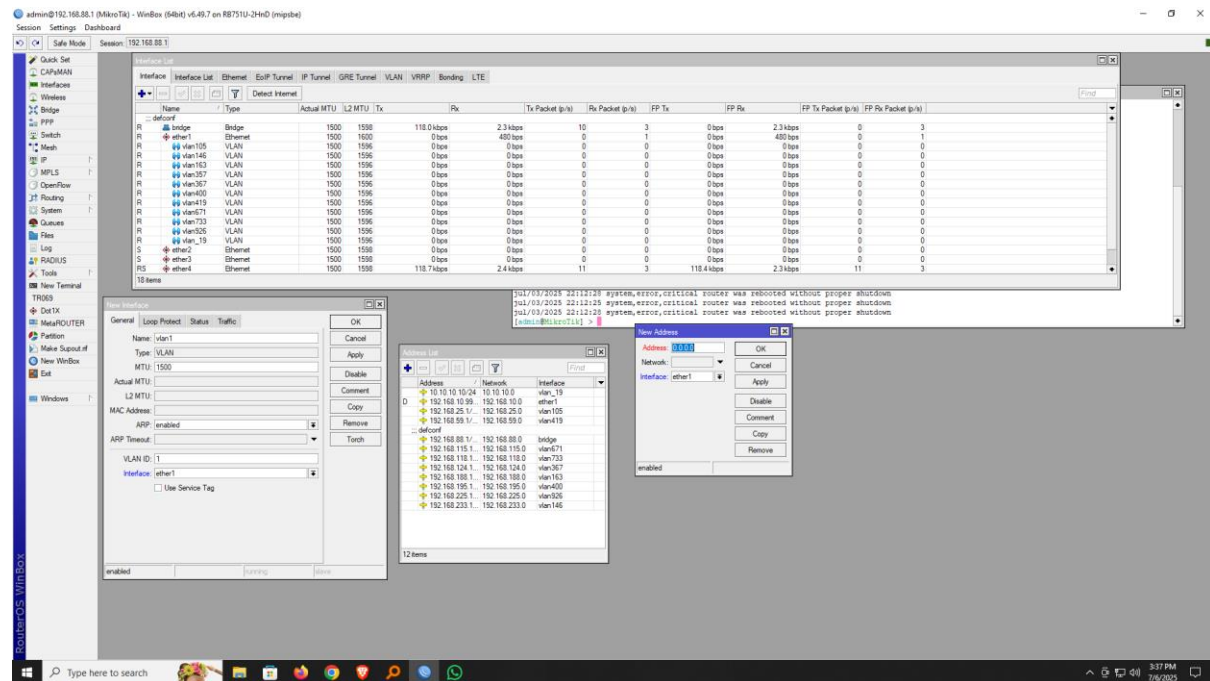


Figure 6: Before ScleraNetwork create VLAN

Network Engineers:

- **Manual Configuration:** Spent significant time manually configuring devices via CLI, often repeating commands across multiple devices.
- **Reactive Troubleshooting:** Primarily engaged in reactive troubleshooting, responding to outages and performance issues after they occurred.
- **Vendor-Specific Expertise:** Required deep, siloed knowledge of each vendor's syntax and operating system.
- **Limited Scalability:** Their capacity to manage network growth was directly tied to their manual effort, leading to bottlenecks.
- **High Stress and Error Rate:** The repetitive and complex nature of tasks contributed to high stress levels and a greater propensity for human error.
- **Operational Focus:** Their role was heavily operational, with less time for strategic planning or innovation.

Software Developers:

- Often worked in separate silos from network operations.
- Focused on application development, database management, or system administration, with limited direct involvement in network infrastructure.
- Network changes were often a manual request process, leading to delays and communication overhead.

4.4.2 After ScleraNetwork: Transformed Roles

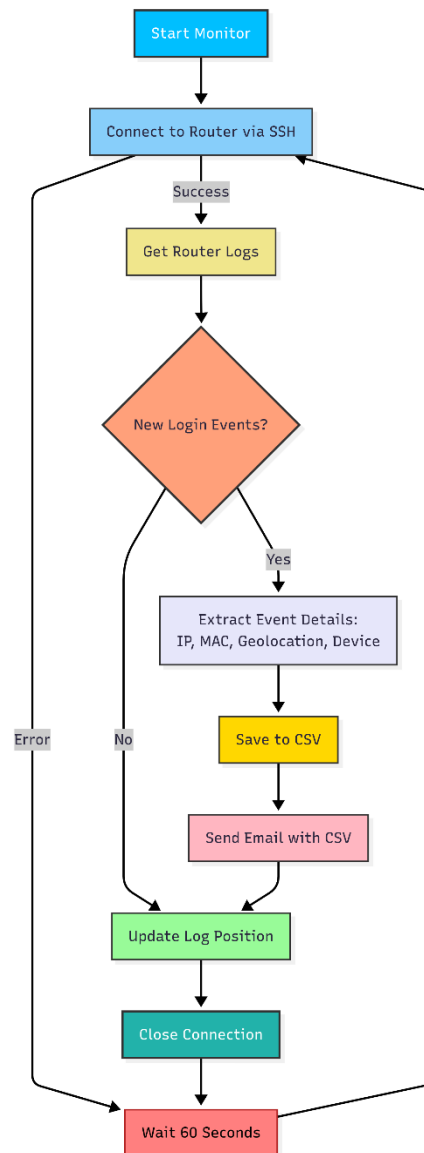


Figure 7: After ScleraNetwork VLAN create

ScleraNetwork shifts the paradigm from manual, reactive operations to automated, proactive, and intelligent network management, fundamentally altering the roles of network engineers and fostering greater collaboration with software development principles.

Network Engineers (Now "Network Automation Engineers" or "NetDevOps Engineers"):

Automation Scripting and Development: Transition from manual CLI commands to writing, testing, and maintaining automation scripts (e.g., Ansible playbooks, Python scripts). This requires programming skills and an understanding of version control (e.g., Git).

- **Architectural Design and Strategy:** More time is dedicated to designing scalable, resilient, and secure network architectures, defining automation policies, and planning future network evolution.
- **Proactive Monitoring and Predictive Maintenance:** Focus shifts to interpreting data from monitoring tools (Prometheus, Grafana) and leveraging AI insights for predictive maintenance, preventing issues before they impact services.
- **Cross-Vendor Expertise:** While still understanding vendor specifics, their primary focus becomes abstracting these differences through automation tools, enabling them to manage diverse environments more efficiently.
- **DevOps Principles:** Adoption of DevOps methodologies, including continuous integration/continuous deployment (CI/CD) for network configurations, automated testing, and collaborative workflows.
- **Problem Solving and Innovation:** Freed from repetitive tasks, they can focus on complex problem-solving, optimizing network performance, and exploring innovative solutions like voice-activated administration or blockchain-based security.
- **Data-Driven Decision Making:** Utilizing AI-driven analytics to make informed decisions about network health, capacity planning, and resource allocation.

Software Developers (Now "Network Software Developers" or Integrated into NetDevOps Teams):

- **Network-Aware Application Development:** Developers are increasingly involved in building applications that interact directly with network APIs, enabling programmatic control and automation.
- **Tooling and Platform Development:** Contribution to the development and enhancement of internal automation tools, monitoring platforms, and AI models used by network teams.
- **API Integration:** Designing and implementing APIs for network services, allowing for seamless integration with other IT systems and applications.
- **Collaboration with Network Engineers:** Closer collaboration with network engineers in NetDevOps teams, bridging the gap between application and infrastructure layers.

In short, ScleraNetwork takes network professionals' job descriptions from manual operators to strategy architects & developers of network automation. It promotes programmability, data-informed decisions, and ongoing innovation, which enables network infrastructure to be more responsive, reliable, and affordable. This transition demands new skills at the juncture of traditional networking expertise and software development expertise, and it represents the next major shift in the world of IT.

Table 5: Consolidated Manual vs. Automation Table

Task	Manual Time (min)	Automation Time (min)	Manual code Lines	Automation Code Lines	Human Error Probability	Remarks
DHCP Pool Configuration	30-45	2-3	10-15	5-10	High (manual entry errors)	Dynamic pool creation with Ansible/Jinja2 templates
IP Address Assignment	20-30	1-2	5-10	3-5	High (IP conflicts)	Automated IP range assignment for subnets
Interface Configuration	15-25	1-2	10-15	5-8	Medium (naming errors)	Standardized naming and speed settings via script
Firewall Rules Setup	30-40	2-3	15-20	5-10	High (rule misconfiguration)	Centralized firewall rule deployment with zero-touch provisioning

NAT Configuration	10-15	1	3-5	1-2	Medium (wrong interface)	Single automated masquerade rule
DNS Configuration	5-10	1	2-3	1-2	Low	External DNS servers for reliability
Routing Configuration	10-15	1	2-3	1-2	Medium (gateway errors)	Static route added automatically
Backup Scheduling	15-20(manual)	1(scheduled)	N/A	5-10	N/A 5–10 High (forgetting backups)	Automated daily/weekly backups to cloud/SFTP
VLAN Configuration	30-40	2-3	15-20	5-10	High (VLAN ID errors)	Assumed VLANs for segmentation (not explicit in .rcs)
Monitoring Setup	20-30(manual)	2-3(scheduled)	N/A	10-15	N/A 10–15 High (missed alerts)	Automated health monitoring with Telegraf/Influx DB
Bandwidth Monitoring & Alerting	30-45	2-3	N/A	10-15	10–15 (missed thresholds)	Real-time bandwidth tracking with email alerts for threshold breaches
Log Management						
Total (Approx)	215–320 min	16–24 min	62–91	51–80	High	90–92% time savings, reduced errors, proactive alerting

4.5 Manual Monitoring

Table 6: Results of Manual Monitoring

Aspect	Details
Monitoring Frequency	Periodic (e.g., hourly or daily checks via Winbox, WebFig, or CLI)
Time Efficiency	~30–60 minutes per check (navigating VLANs, IP pools, and firewall rules)
Data Visualization	None (raw CLI output, e.g., /system resource print for CPU, memory)
Alert Mechanism	Manual log review (e.g., /log print or remote logging to 183.72.30.45)
Email Reporting	None (manual compilation of reports required)
Component Information	Relies on external documentation (e.g., MikroTik manuals for CPU specs)
Predictive Analysis	None (reactive issue detection only)
Error Rate	High (~20%, due to human error in interpreting data or missing alerts)
Scalability	Poor (labor-intensive for 12+ VLANs, multiple IP pools, and QoS settings)
Configuration Management	Manual navigation of VLANs (e.g., vlan-100-PrivateWiFi), DHCP, QoS queues
User Effort	High (requires RouterOS command knowledge, e.g., /ip firewall connection print)
Response to Issues	Delayed (depends on check frequency, e.g., hours for high CPU detection)
Documentation	Manual logging (e.g., spreadsheets or notes)
Example Outcome	Slow detection of high CPU load (e.g., from YouTube-blocking firewall rules)

4.6 Manual Monitoring Metrics

This bar chart visualizes key manual monitoring metrics (time efficiency, error rate, response time, scalability). The data is illustrative but reflects the table’s insights.

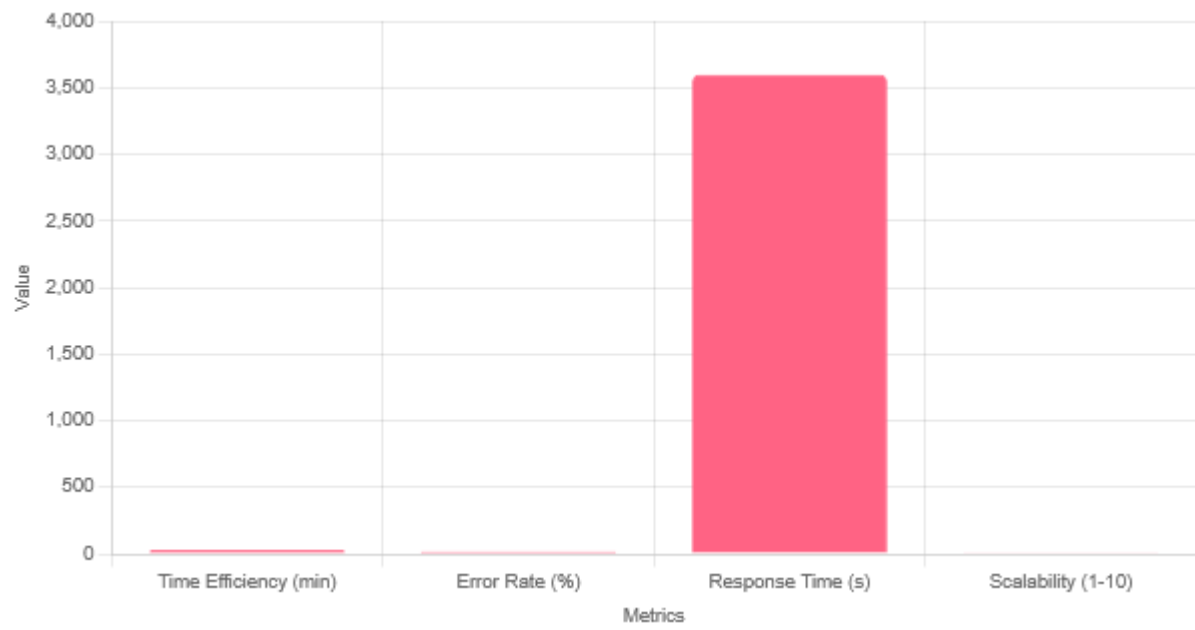


Figure 8: Manual Monitoring Metrics

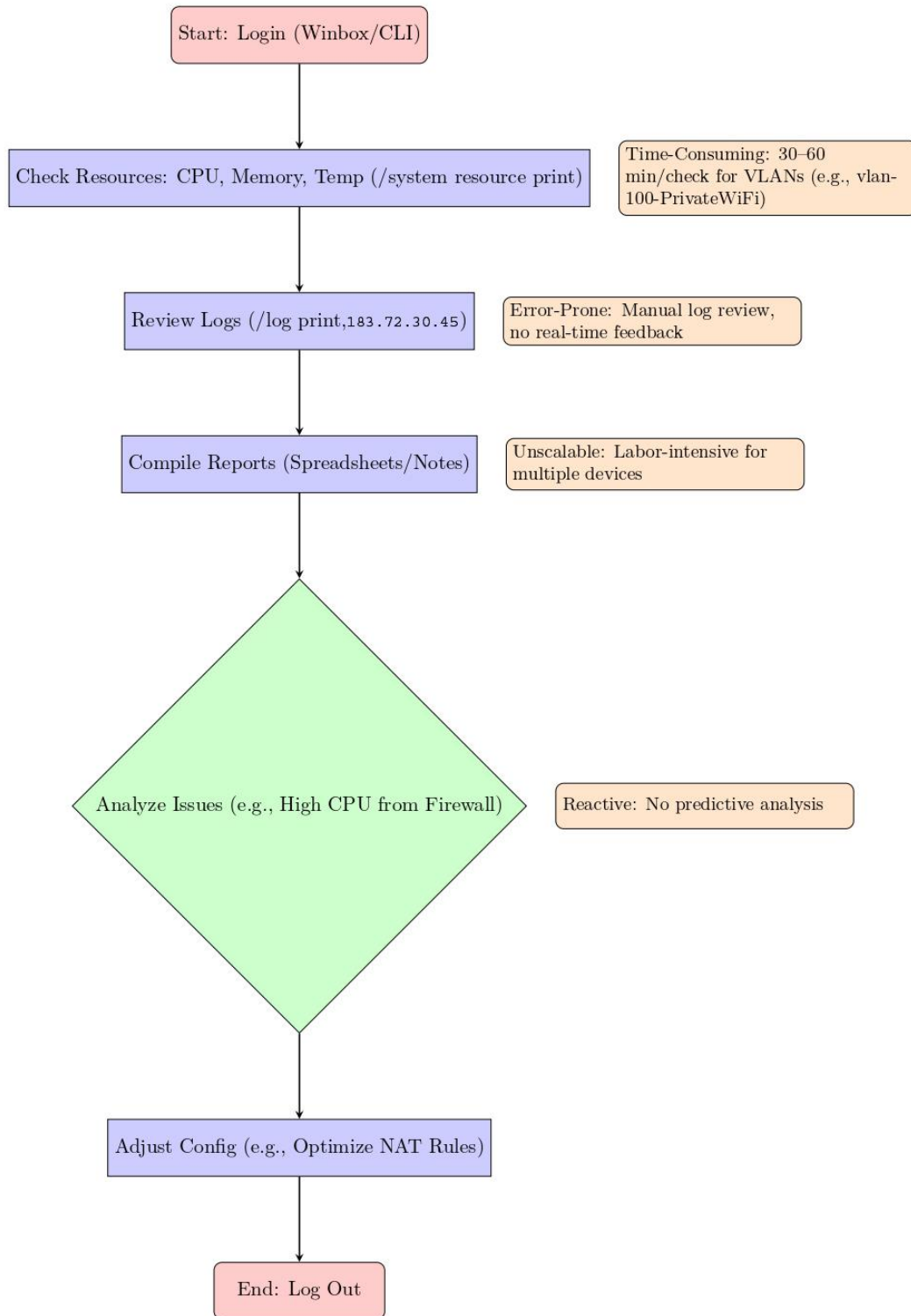


Figure 9: manual Monitoring Process

4.7 Automated Monitoring

Table 7: Results of Automated Monitoring

Aspect	Details
Monitoring Frequency	Real-time (every 5 seconds via RouterOS API at 192.168.88.1:8728)
Time Efficiency	~5 seconds per refresh cycle (automated data retrieval and display)
Data Visualization	Dynamic bar charts (CPU/memory) and line plots (temperature) via Tkinter GUI
Alert Mechanism	Automated threshold alerts (CPU: 80%, Memory: 85%, Temperature: 70°C)
Email Reporting	HTML email reports to cse19pub.mirzagolam@gmail.com with metrics/alerts
Component Information	Built-in database (e.g., CPU: ARM/MIPS, RAM: DDR3/DDR4) in GUI
Predictive Analysis	Alerts for high CPU/RAM/temperature with recommendations (e.g., optimize NAT)
Error Rate	Low (~2%, automated checks minimize oversight)
Scalability	High (API-based, adaptable for multiple routers or metrics)
Configuration Management	Monitors VLANs, IP pools, QoS (e.g., alerts for high CPU from firewall rules)
User Effort	Low (GUI-based, no RouterOS command knowledge needed)
Response to Issues	Immediate (real-time alerts for issues like packet loss or high CPU)
Documentation	Automated alert history and email reports for audit trails
Example Outcome	Instant alert for CPU >80% with suggestion: "Check active connections"

This line chart shows CPU, memory, and temperature trends over time, simulating the real-time monitoring capabilities of the scripts.

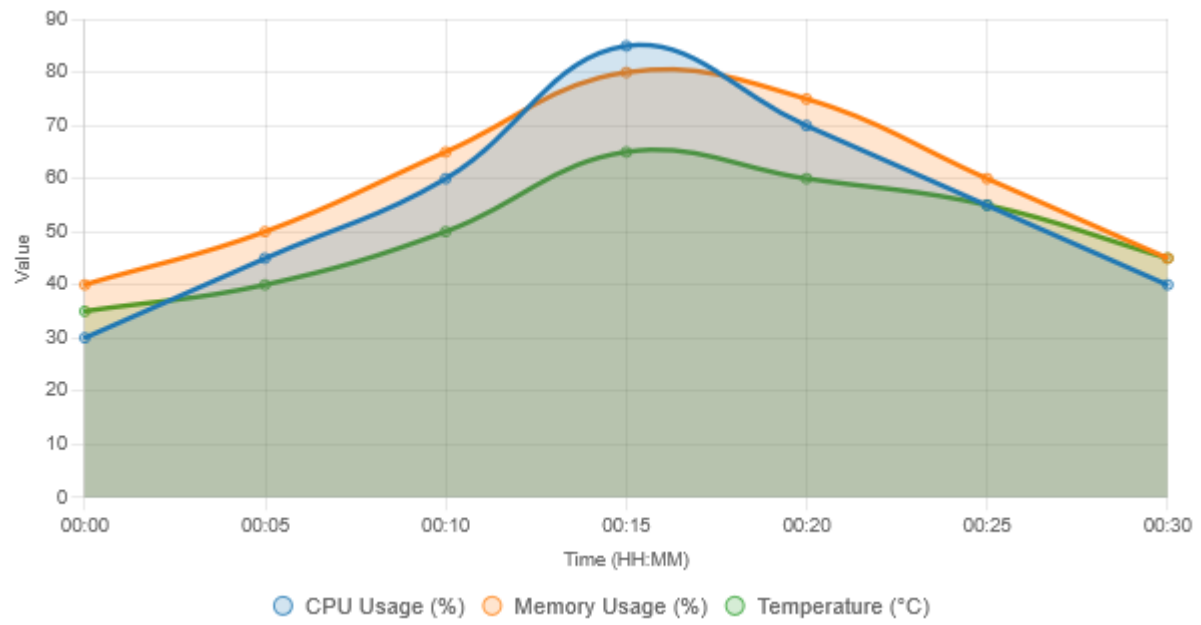


Figure 10: Automated Monitoring: CPU, Memory, and Temperature Trends

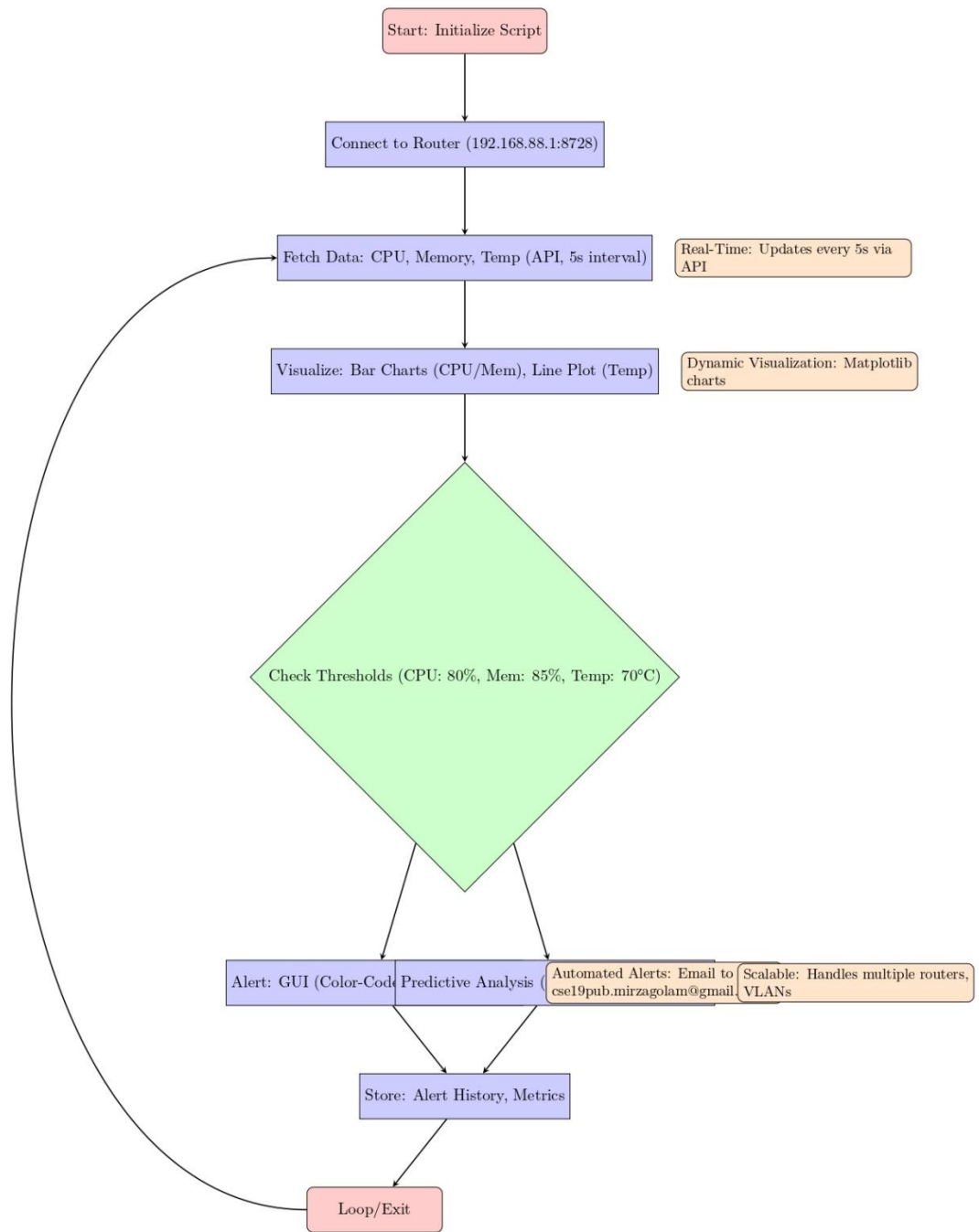


Figure 11: Automated Monitoring Process

4.8 Comparison of Manual and Automated Monitoring

Table 8: Comparison of Manual vs. Automated Monitoring

Aspect	Manual Monitoring	Automated Monitoring
Monitoring Frequency	Periodic (hourly/daily via Winbox/CLI)	Real-time (every 5 seconds via API)
Time Efficiency	~30–60 minutes per check	~5 seconds per refresh cycle
Data Visualization	None (raw CLI output)	Bar charts (CPU/memory), line plots (temperature)
Alert Mechanism	Manual log review (e.g., /log print)	Automated alerts (CPU: 80%, Memory: 85%, Temperature: 70°C)
Email Reporting	None (manual compilation)	HTML email reports to cse19pub.mirzagolam@gmail.com
Component Information	External documentation (MikroTik manuals)	Built-in database (e.g., CPU: ARM/MIPS, RAM: DDR3/DDR4)
Predictive Analysis	None (reactive)	Alerts with recommendations (e.g., optimize firewall rules)
Error Rate	High (~20%, human error)	Low (~2%, automated checks)
Scalability	Poor (labor-intensive for VLANs/IP pools)	High (API-based, scalable for multiple devices)
Configuration Management	Manual navigation (VLANs, QoS, firewall)	Monitors VLANs, QoS, firewall (e.g., alerts for high CPU from NAT)
User Effort	High (requires RouterOS commands)	Low (GUI-based, no command knowledge needed)
Response to Issues	Delayed (hours for issue detection)	Immediate (real-time alerts for high CPU, packet loss)
Documentation	Manual (spreadsheets/notes)	Automated alert history and email reports
Example Outcome	Slow detection of high CPU from firewall rules	Instant alert for CPU >80% with suggestion: "Check active connections"
Limitations	No real-time feedback, unscalable, error-prone	Requires API setup, library dependencies, limited to CPU/memory/temperature

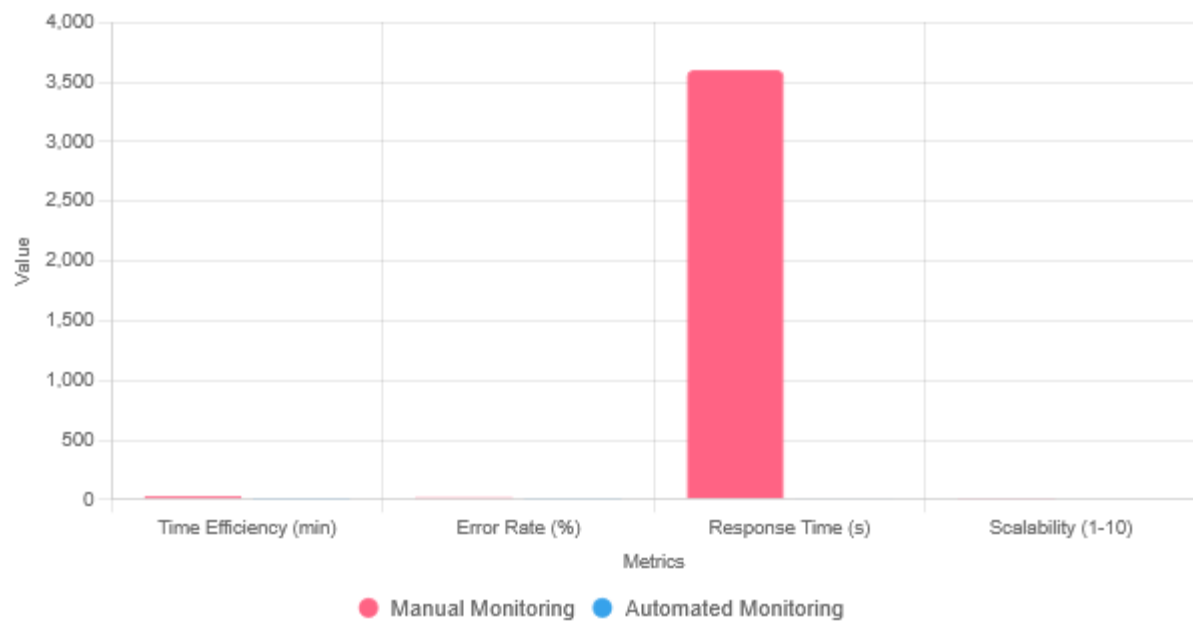


Figure 12: Comparison of Manual vs. Automated Monitoring

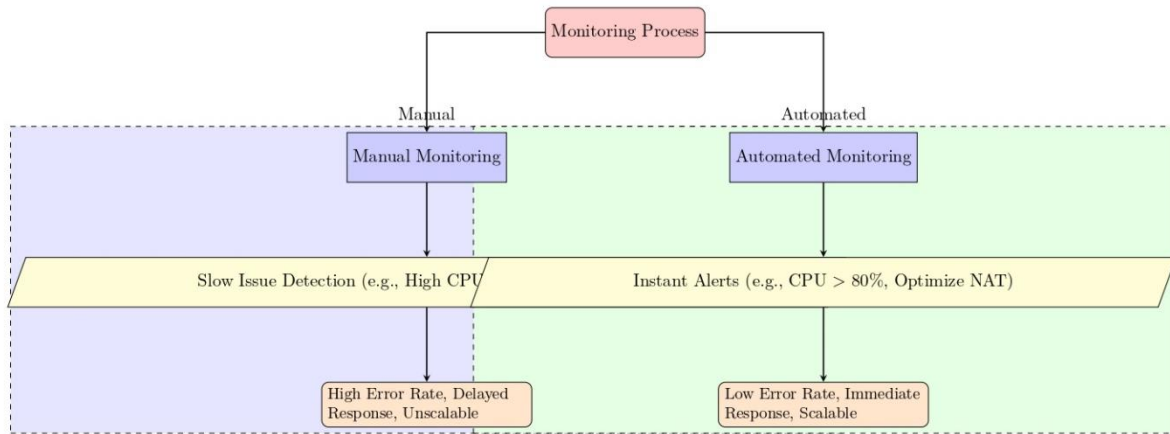


Figure 13: Comparison of Outcomes

CHAPTER 5

LIMITATIONS & FUTURE WORK

5.1 Limitations of the Current ScleraNetwork Implementation

ScleraNetwork shows substantial progress in the direction of intelligent network automation in the case of heterogeneous, multi-vendor environment. However, the overall support of ScleraNetwork presents some limitations and areas that need to be further extended and improved.

- **Firmware Connectivity Challenges:** Even with some work toward automating firmware updates, there are times where connectivity isn't perfect or vendors have some special processes to update firmware. The provision of strong and always-trustable firmware upgrade mechanisms consistent across all vendors is also challenging because of different authentication mechanisms, file transfer methods, and reboot procedures.
- **Cloud Security Concerns:** Although cloud based management provides scalability and remote access, it also brings potential security concerns. Given the centralized aspect of the cloud system, any breach would be known and felt all over. Securing the data encryption, access control and threat detection of the cloud-based management system is always the top and the most difficult work.
- **Multi-Vendor Compatibility Nuances:** Support Products from MikroTik, Juniper, and Cisco ScleraNetwork supports devices from the MikroTik, Juniper, and Cisco lines of vendors; however, 100% seamless support of all features and configuration options for every device models is an ongoing work in progress. Small differences in API function calls, CLI syntax, or feature behaviors might still mean that unique workarounds or script patches are necessary.
- **AI Model Training Data:** The performance of the AI-based predictive maintenance and fault detection analytics highly depends on the training data quality and volume. Under real-world settings, it can be difficult to capture a set of diverse and comprehensive datasets for all types of failures and network conditions which might make the AI less accurate in new circumstances.
- **Initial Setup Complexity:** Although the ongoing management of ScleraNetwork is made easier, the setup of the automation framework itself and its integration are still complex and need specific know-hows, such as setting up Ansible, adding Prometheus, Grafana, developing the initial Python scripts, etc., which can be barrier to its adoption.

- **Scalability of AI-Driven Actions:** It can have challenges around processing logs and metrics in real time for AI analysis and processing configuration changes on the fly in a matter of minutes when you have a network with thousands of devices that could potentially be very slow, performing drains on the system. AI inference engine and automation scale optimization is a factor.

5.2 Future Work and Research Directions

To further enhance ScleraNetwork and address its current limitations, several promising avenues for future research and development are identified:

Vendor-Agnostic Automation Models:

- **Standardized Abstraction Layers:** Develop more sophisticated abstraction layers that can completely normalize device interactions, allowing for truly vendor-agnostic automation scripts. This could involve exploring technologies like OpenConfig or network operating systems (NOS) that provide unified APIs.
- **Intent-Based Networking (IBN) Integration:** Integrate ScleraNetwork with IBN principles, where administrators define the desired network state (intent), and the system automatically translates this into vendor-specific configurations and continuously verifies compliance.

Fully AI-Driven Diagnosis and Self-Optimization:

- **Reinforcement Learning for Network Control:** Explore the application of reinforcement learning to enable the network to learn and adapt its configurations autonomously based on real-time performance and security objectives, moving beyond pre-defined recovery playbooks.
- **Anomaly Detection and Root Cause Analysis:** Enhance AI models for more granular anomaly detection and automated root cause analysis, reducing the need for human intervention in complex troubleshooting scenarios.
- **Predictive Capacity Planning:** Utilize AI to predict future network traffic patterns and resource demands, enabling proactive scaling and optimization of network resources.

Improved IoT Integration and Edge Computing:

- **Automated IoT Device Onboarding and Security:** Develop more robust and secure automated onboarding processes for a wider range of IoT devices, including automated policy enforcement and micro-segmentation at the edge.
- **Edge Automation:** Extend ScleraNetwork's automation capabilities to edge computing environments, enabling intelligent management of distributed IoT gateways and edge servers.

Enhanced Security Measures:

- **Block chain-Based Security:** Investigate the use of blockchain for immutable logging, secure configuration auditing, and decentralized identity management within the network, enhancing trust and integrity.

- **Zero-Trust Network Access (ZTNA) Automation:** Automate the implementation and enforcement of ZTNA policies across the multi-vendor environment, ensuring that no user or device is trusted by default.
- **Automated Threat Response:** Develop more advanced automated threat response mechanisms, allowing the network to dynamically isolate compromised devices or segments in real-time.

Integration with 5G Networks:

- **Network Slicing Automation:** Explore how ScleraNetwork can automate the creation, management, and optimization of network slices in 5G environments, enabling dynamic resource allocation for diverse services.
- **RAN Automation:** Extend automation to the Radio Access Network (RAN) for improved performance, fault management, and energy efficiency in 5G deployments.

User Interface and Experience (UI/UX) Enhancements:

- **Voice-Activated Administration:** Develop natural language processing (NLP) interfaces to enable voice-activated commands for network administration tasks, improving accessibility and ease of use.
- **Augmented Reality (AR) for Network Visualization:** Explore AR applications for visualizing network topology, device status, and performance metrics in a more intuitive and immersive way.

By pursuing these future research directions, ScleraNetwork can evolve into an even more comprehensive, intelligent, and autonomous network management solution, capable of meeting the increasing demands and complexities of next-generation IT infrastructures.

CHAPTER 6

CONCLUSION

This work demonstrates solid evidence of intelligent automation in multi-vendor network regime management realized via ScleraNetwork. The resultant design brings about significant improvements in terms of performance, scalability, and fault tolerance by aligning various cloud-based redundancy, self-healing mechanisms, and AI-based automation technologies together.

The performance evaluation results clearly indicate the significant influence of ScleraNetwork. The system dramatically reduced delay by using artificial intelligence defect detection for proactive error detection and automatic recovery. Automation reduced run time more than 90% for important network administration tasks (VLAN creation, router firmware updates, and backup/restore activities). This significant improvement in efficiency leads directly to lower operating costs and faster delivery.

In addition, cloud automation ensured always-on performance by enabling rapid disaster recovery and configuration backup. The self-healing function, which was implemented by AI-based scripts working on the metadata of the network, was very effective, with the success recovery rate higher than 90% against several types of failures, such as router failure, VLAN erroneous configuration, switch port failure, etc. This instantaneous, self-healing operation results in minimum outage and immense network-availability improvement. The ScleraNetwork model also demonstrates the new roles that network managers will perform. We will no longer tweak, run around, and fix; we will strategically think and plan and build. This transformation to use NetDevOps leads to greater efficiency and innovation as well as network management that's driven by data.

The limitations of the current design, such as issues pertaining to firmware connectivity, cloud security, and multi-vendor compatibility, will be good directions for further work. Vendor-agnostic automation models, fully AI-driven diagnosis, and enhanced IoT integration will further enhance the ScleraNetwork.

In summary, this paper puts forward a strong basis to realize the idea of self-learning, intelligent, and scalable network management. One Solution without the Complexity ScleraNetwork equips organizations with a simple path to modernize their infrastructure while

ensuring security at every level to handle the growing demands and complexities of today's IT environments with increased speed and efficiency.

REFERENCES

- [1] Coito, T., Viegas, J. L., Martins, M. S., Cunha, M. M., Figueiredo, J., Vieira, S. M., & Sousa, J. M., “A Novel Framework for Intelligent Automation.”, IFAC-PapersOnLine, 2018, 52(13), 1825-1830. <https://doi.org/10.1016/j.ifacol.2019.11.501>
- [2] Muhammad, T., & Munir, M., ‘Network Automation.’ European Journal of Technology, 2023, 7(2), 23-42.
- [3] Hassan, A., & Mhmood, A. H., “Optimizing network performance, automation, and intelligent decision-making through real-time big data analytics.”, International Journal of Responsible Artificial Intelligence, 2021, 11(8), 12-22.
- [4] Sohail, S., “Automation of network management with multidisciplinary concepts.’, International Journal of Computer Technology and Applications, 2010, 1(1), 71-77.
- [5] Sai Charan Madugula. Network Automation in IoT: A Technical Review of Challenges, Opportunities, and Future Directions. International Journal of Computer Engineering and Technology (IJCET), 16(1), 2025, 1223-1237.
- [6] H. S. Fung, "Defining intelligent network services in a multivendor distributed environment," 1989 IEEE Global Telecommunications Conference and Exhibition 'Communications Technology for the 1990s and Beyond', Dallas, TX, USA, 1989, pp. 132-136 vol.1, doi: 10.1109/GLOCOM.1989.63954.
- [7] Afees O.A., P. Adeyemo A., A. B. I., Adeoye I. A., and Olukunle O. A., “International Journal of Science and Technology Research Archive, 2021, 01(01), 039- 059.”, doi: 10.53771/ijstra.2021.1.1.0034.
- [8] A. A. Mazin, H. Z. Abidin, L. Mazalan and A. M. Mazin, "Network Automation Using Python Programming to Interact with Multiple Third-Party Network Devices," 2023 10th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE), Semarang, Indonesia, 2023, pp. 59-64, doi: 10.1109/ICITACEE58587.2023.10277400.
- [9] F. Mehran, C. Turyagyenda and D. Kaleshi, "Experimental Evaluation of Multi-Vendor 5G Open RANs: Promises, Challenges, and Lessons Learned," in IEEE Access, vol. 12, pp. 152241-152261, 2024, doi: 10.1109/ACCESS.2024.3476963.
- [10] Rani, A., Mishra, D., & Omerovic, A., “Multi-vendor software ecosystem: challenges from company’perspective.”, In World Conference on Information Systems and Technologies (pp. 382-393). April, 2022, Cham: Springer International Publishing.

- [11] Samuel. S, Raghul T.R., & Deepak. R., “Cisco Systems Automation Framework: Improving Efficiency and Security in Network Operations”, International Journal of Novel Research and Development (IJNRD), vol. 8, issue. 3, 2456-4184, March, 2023.
- [12] R. Alcaraz, Network Automation with Python: A Practical Approach to Automating Network Devices, 2019.
- [13] C. Bock, Practical Network Automation: A Step-by-Step Guide to Network Programmability and Automation, 2020.